

2023 AMC 10B Problem 8 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 8. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 8

Problem:

What is the units digit of $2022^{2023} + 2023^{2022}$?

Answer Choices:

- A. 7
- B. 1
- C. 9
- D. 5
- E. 3

Solution:

The units digit of 2022^{2023} is the same as the units digit of 2^{2023} . The units digit of the powers of 2 cycle: 2, 4, 8, 6, 2, 4, Because 2023 has remainder 3 when divided by 4, the units digit of 2^{2023} is 8. Similarly, the units digit of 2023^{2022} is the same as the units digit of 3^{2022} . The units digit of the powers of 3 cycle: 3, 9, 7, 1, 3, 9, Because 2022 has remainder 2 when divided by 4, the units digit of 3^{2022} is 9. Thus the units digit of $2022^{2023} + 2023^{2022}$ is the same as the units digit of $8 + 9 = 17$, which is 7.

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